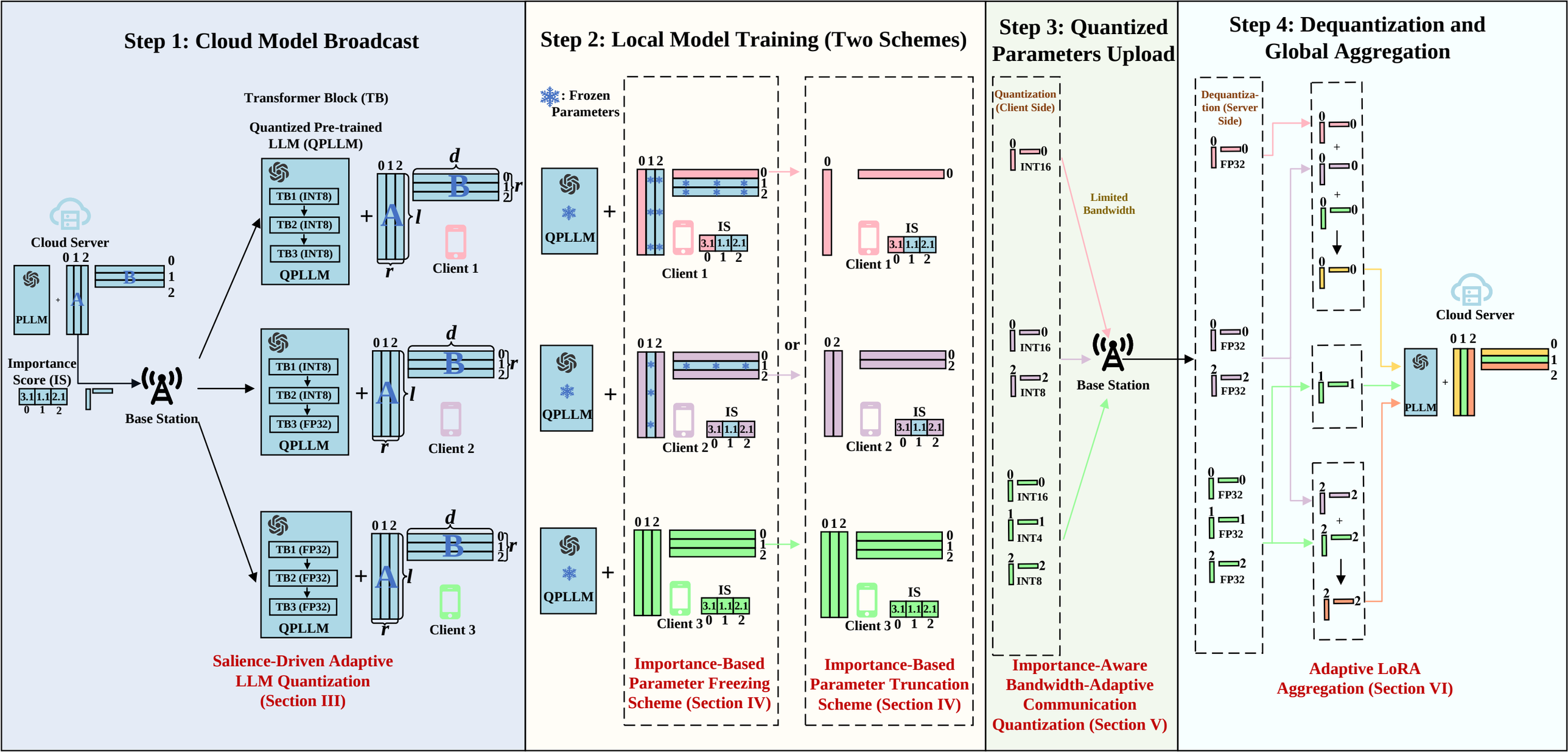

Step 1: Cloud Model Broadcast
Transformer Block (TB)
Quantized Pre-trained LLM (QPLLM)
TB1 (INT8)
TB2 (INT8)
TB3 (INT8)
QPLLM
TB1 (INT8)
TB2 (INT8)
TB3 (FP32)
QPLLM
TB1 (FP32)
TB2 (FP32)
TB3 (FP32)
QPLLM
Cloud Server
PLLM
Importance Score (IS)
Base Station
Client 1
Client 2
Client 3
Salience-Driven Adaptive LLM Quantization (Section III)
Step 2: Local Model Training (Two Schemes)
: Frozen Parameters
QPLLM
Client 1
Client 2
Client 3
or
Importance-Based Parameter Freezing Scheme (Section IV)
Importance-Based Parameter Truncation Scheme (Section IV)
Step 3: Quantized Parameters Upload
Quantization (Client Side)
INT16
INT8
INT4
Limited Bandwidth
Base Station
Importance-Aware Bandwidth-Adaptive Communication Quantization (Section V)
Step 4: Dequantization and Global Aggregation
Dequantiza-tion (Server Side)
FP32
Cloud Server
PLLM
Adaptive LoRA Aggregation (Section VI)